

8. (a) When a mixture of iron(III) oxide and aluminium powder (Thermit mixture) is heated, there is a violent reaction. The reaction is carried out in a tube surrounded by a mound of sand because the temperature reaches 2500 °C. A bead of iron is recovered from the sand. The picture below shows the reaction taking place in a darkened room.



- (i) Give the reason why the iron formed in the reaction is molten. [1]

.....
.....

- (ii) Complete and balance the **symbol** equation for this reaction. [2]



- (iii) State which of the substances is oxidised. Give the reason for your choice. [1]

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.....

- (iv) When a mixture of magnesium oxide and aluminium powder is heated, there is no reaction.

List iron, magnesium and aluminium in order of reactivity. [1]

Most reactive

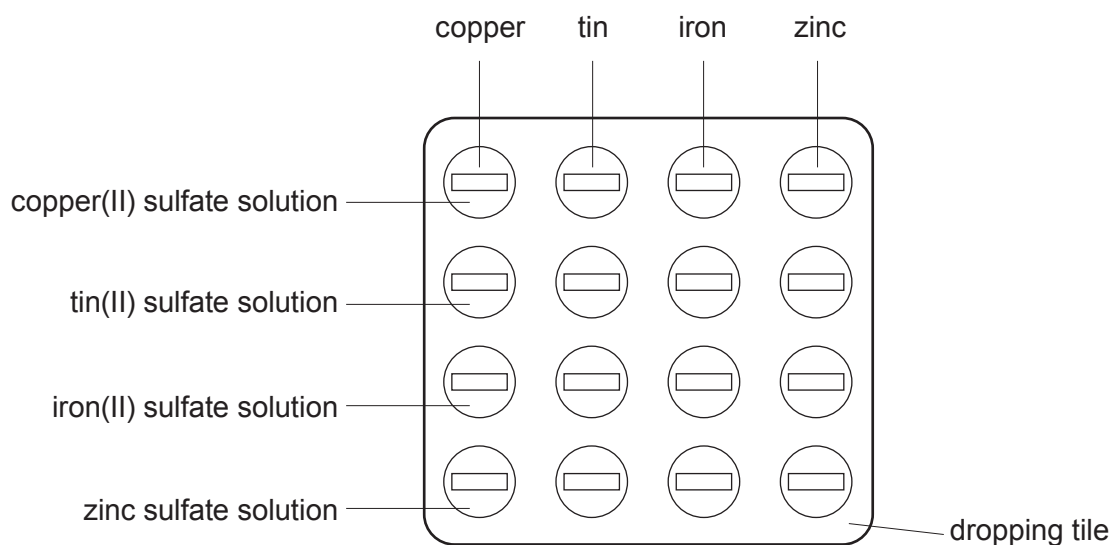
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Least reactive



- (b) Some metals are more reactive than others. A more reactive metal displaces a less reactive metal from its compounds.

A student was given tin, iron, copper and zinc and solutions of the metal sulfates. Using a dropping pipette, she put a little of one of the sulfate solutions in four of the depressions of the dropping tile. She did this for each solution in turn. She then put a piece of metal foil in each of the solutions, as shown below.



- (i) Put a tick (✓) next to the question which **best** describes the investigation the student is carrying out. [1]

Which displacement is the most exothermic?

☐

Which metal can displace copper from solution?

☐

What is meant by the reactivity series?

☐

What are the positions of the four metals in the reactivity series?

☐

- (ii) The student recorded the results by putting a tick (✓) next to a mixture which showed signs of a reaction and a cross (X) next to a mixture which showed no signs of a reaction.

The student concluded that:

tin displaces copper
iron displaces tin
iron displaces copper
zinc displaces iron

Give the **letter** of the tile below which shows the results she recorded.

[1]

Letter

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

A

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

B

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

C

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

D

- (iii) Another student said that not all of the tests were necessary. Give **one** example of a test not needed. Explain your choice.

[2]

Example

Explanation

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- (c) Copper displaces silver from a solution of silver nitrate, AgNO_3 , to form copper(II) nitrate solution.

(i) Describe **one** change the student would **see** during this displacement reaction. [1]

.....

.....

(ii) Write a balanced **symbol** equation for this reaction. [2]

.....

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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0

1 H Hydrogen 1										4 He Helium 2																							
7 Li Lithium 3	9 Be Beryllium 4		19 F Fluorine 9		16 O Oxygen 8		14 N Nitrogen 7		12 C Carbon 6		11 B Boron 5		20 Ne Neon 10		35.5 Cl Chlorine 17		40 Ar Argon 18																
	23 Na Sodium 11		24 Mg Magnesium 12		32 S Sulfur 16		31 P Phosphorus 15		28 Si Silicon 14		27 Al Aluminium 13		84 Kr Krypton 36		127 I Iodine 53		131 Xe Xenon 54																
39 K Potassium 19	40 Ca Calcium 20		59 Ni Nickel 28		63.5 Cu Copper 29		59 Co Cobalt 27		73 Ge Germanium 32		70 Ga Gallium 31		79 Se Selenium 34		80 Br Bromine 35		84 Kr Krypton 36																
	86 Rb Rubidium 37		88 Sr Strontium 38		108 Ag Silver 47		103 Rh Rhodium 45		119 Sn Tin 50		115 In Indium 49		128 Te Tellurium 52		127 I Iodine 53		131 Xe Xenon 54																
133 Cs Caesium 55	137 Ba Barium 56		181 Ta Tantalum 73		186 Re Rhenium 75		192 Ir Iridium 77		207 Pb Lead 82		204 Tl Thallium 81		210 Po Polonium 84		210 At Astatine 85		222 Rn Radon 86																
	223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		65 Zn Zinc 30		75 As Arsenic 33																		